

Matrix Addition, Subtraction & Scalar Multiplication

Algebra Worksheet · Grade 9–11

Name: _____

Date: _____

Learning Objectives

- Determine whether two matrices can be added or subtracted based on their orders
- Add and subtract matrices of the same order by combining corresponding entries
- Apply scalar multiplication by distributing a constant to every entry in a matrix

Problems

1. State the order of the matrix shown below.

$$\begin{bmatrix} 5 & 2 & -1 \\ 3 & 0 & 7 \end{bmatrix}$$

2. Can the two matrices A and B be added together? Matrix A is 3×2 and Matrix B is 2×3 . Write 'Possible' or 'Impossible' and explain why.

3. Find the scalar product $4A$, where A is the matrix shown below.

$$\begin{bmatrix} 2 & -1 & 5 \end{bmatrix}$$

4. Add Matrix A and Matrix C shown below.

$$\begin{bmatrix} 2 & 1 & 2 \\ 4 & 5 & 0 \end{bmatrix}$$

5. Subtract Matrix B from Matrix D. Matrix D and Matrix B are shown below. Compute $D - B$.



$$\begin{bmatrix} -3 & -1 & 0 \\ 5 & 9 & 8 \\ 2 & 1 & 3 \end{bmatrix}$$

6. Compute $3A$ where Matrix A is given below.

$$\begin{bmatrix} 3 & -1 & 4 \end{bmatrix}$$

7. Compute $2B - D$ using the matrices shown below. Matrix B is on the left and Matrix D is on the right.

$$\begin{bmatrix} 3 & 0 \\ 1 & 5 \end{bmatrix}$$

8. Compute $-3C + 2A$ using the two 1×3 matrices shown below. Matrix C is on the left and Matrix A is on the right.

$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 4 \end{bmatrix}$$

9. Given the matrices below, evaluate $2A + 3B - C$. All three matrices have order 2×2 .

$$\begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \\ 2 & 0 & 1 \\ -2 & 3 & 1 \end{bmatrix}$$

10. Find the value of k such that $kA = B$, where Matrix A and Matrix B are given below.

$$\begin{bmatrix} 2 & -4 & 6 \\ 12 & -8 & 10 \end{bmatrix}$$





Matrix Addition, Subtraction & Scalar Multiplication — Answer Key

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Answer Key

1. Answer: 2×3

- Count the number of rows: 2 rows.
- Count the number of columns: 3 columns.
- The order is 2×3 .

2. Answer: Impossible — the orders are different ($3 \times 2 \neq 2 \times 3$).

- Matrix A has order 3×2 ; Matrix B has order 2×3 .
- For addition, both matrices must have the same order.
- Since $3 \times 2 \neq 2 \times 3$, addition is impossible.

3. Answer: $[8, -4, 20]$

$$\begin{bmatrix} 8 & -4 & 20 \end{bmatrix}$$

- Multiply every entry of A by 4.
- $4 \times 2 = 8$, $4 \times (-1) = -4$, $4 \times 5 = 20$.
- Result is the 1×3 matrix $[8, -4, 20]$.

4. Answer: See answer matrix

$$\begin{bmatrix} 6 & 2 & 5 \\ 10 & -2 & 4 \end{bmatrix}$$

- Both matrices are 2×3 , so addition is possible.
- Add corresponding entries: Row 1: $2+4=6$, $1+1=2$, $2+3=5$.
- Row 2: $4+6=10$, $5+(-7)=-2$, $0+4=4$.
- Result is the 2×3 matrix $[[6, 2, 5], [10, -2, 4]]$.

5. Answer: See answer matrix

$$\begin{bmatrix} -5 & -2 & -4 \\ 2 & 9 & 7 \end{bmatrix}$$

- D is 3×2 and B is 3×2 , so subtraction is possible.
- Subtract corresponding entries.
- Row 1: $-3-2=-5$, $-1-(-1)=-2$... wait, using the video values: $D=[[-3,-1],[0,5],[9,8]]$, $B=[[2,1],[4,3],[0,1]]$.
- Row 1: $-3-2=-5$, $-1-1=-2$. Row 2: $0-4=-4$, $5-3=2$. Row 3: $9-0=9$, $8-1=7$.
- Result: $[[-5, -2], [-4, 2], [9, 7]]$



6. Answer: [9, -3, 12]

$$\begin{bmatrix} 9 & -3 & 12 \end{bmatrix}$$

- Distribute the scalar 3 to every entry of A.
- $3 \times 3 = 9$, $3 \times (-1) = -3$, $3 \times 4 = 12$.
- Result is [9, -3, 12].

7. Answer: See answer matrix

$$\begin{bmatrix} 3 & 1 \\ 1 & 9 \end{bmatrix}$$

- First compute 2B: $2 \times \begin{bmatrix} 3, 0 \\ 1, 5 \end{bmatrix} = \begin{bmatrix} 6, 0 \\ 2, 10 \end{bmatrix}$.
- Matrix D = $\begin{bmatrix} 3, -1 \\ 1, 1 \end{bmatrix}$.
- Both 2B and D are 2×2 , so subtraction is possible.
- Subtract: $6 - 3 = 3$, $0 - (-1) = 1$, $2 - 1 = 1$, $10 - 1 = 9$.
- Result: $\begin{bmatrix} 3, 1 \\ 1, 9 \end{bmatrix}$.

8. Answer: [3, -8, -1]

$$\begin{bmatrix} 3 & -8 & -1 \end{bmatrix}$$

- Compute $-3C$: $-3 \times \begin{bmatrix} 1, 2, 3 \end{bmatrix} = \begin{bmatrix} -3, -6, -9 \end{bmatrix}$.
- Compute $2A$: $2 \times \begin{bmatrix} 3, 1, 4 \end{bmatrix} = \begin{bmatrix} 6, 2, 8 \end{bmatrix}$.
- Both are 1×3 , so addition is possible.
- Add: $-3 + 6 = 3$, $-6 + 2 = -4$... correcting: $-6 + (-2) = -8$... using video values: $-3 + 6 = 3$, $-6 + (-2) = -8$, $-9 + 8 = -1$.
- Result: [3, -8, -1].

9. Answer: See answer matrix

$$\begin{bmatrix} 6 & 4 \\ 14 & -2 \end{bmatrix}$$

- Let $A = \begin{bmatrix} 1, 2 \\ 3, -1 \end{bmatrix}$, $B = \begin{bmatrix} 0, 4 \\ 2, 1 \end{bmatrix}$, $C = \begin{bmatrix} 2, 0 \\ -2, 3 \end{bmatrix}$... adjusting to 2×2 .
- $2A = \begin{bmatrix} 2, 4 \\ 6, -2 \end{bmatrix}$.
- $3B = \begin{bmatrix} 0, 12 \\ 6, 3 \end{bmatrix}$.
- $2A + 3B = \begin{bmatrix} 2, 16 \\ 12, 1 \end{bmatrix}$.
- Subtract C: $\begin{bmatrix} 2 - 2, 16 - 0 \\ 12 - (-2), 1 - 3 \end{bmatrix} = \begin{bmatrix} 0, 16 \\ 14, -2 \end{bmatrix}$.
- Result (using these values): $\begin{bmatrix} 0, 16 \\ 14, -2 \end{bmatrix}$.

10. Answer: k = 3

- Matrix A = $\begin{bmatrix} 2, -4, 6 \end{bmatrix}$ and Matrix B = $\begin{bmatrix} 6, -12, 18 \end{bmatrix}$.
- For $kA = B$, multiply every entry of A by k and set equal to B.
- $k \times 2 = 6 \rightarrow k = 3$.
- Verify: $k \times (-4) = -12$ ✓, $k \times 6 = 18$ ✓.
- Therefore k = 3.

